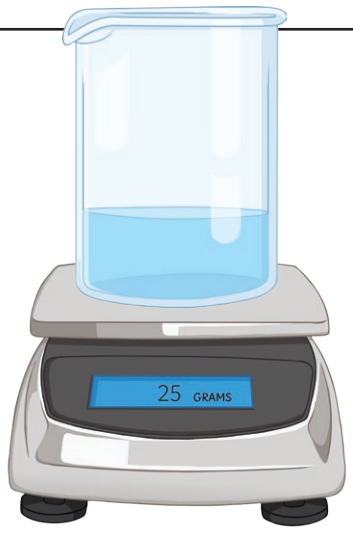
Conservation of Mass

The law of the conservation of mass states that the total mass of the reactants is equal to the total mass of the products.

Complete the table below to explain why it doesn't always appear that the mass of the reactants equals the mass of the products.

Experiment	What happens to the mass?	Explanation
10g of magnesium ribbon is burnt in oxygen		
25g of hydrochloric acid has 3g of powdered marble added to it, a gas escapes		
25g of water has 5g of salt added to it and is stirred until all the salt dissolves		
A can of lemonade is opened and left for 4 hours		
An ice cube is placed in a beaker and left to melt		



Conservation of Mass **Answers**

The law of the conservation of mass states that the total mass of the reactants is equal to the total mass of the products.

Complete the table below to explain why it doesn't always appear that the mass of the reactants equals the mass of the products.

Experiment	What happens to the mass?	Explanation
10g of magnesium ribbon is burnt in oxygen	**The mass increases**	**The final mass will be the mass of the magnesium and the mass of the oxygen**
25g of hydrochloric acid has 3g of powdered marble added to it, a gas escapes	**The mass appears to decrease**	**The gas has escaped into the atmosphere and its mass will no longer being measure**
25g of water has 5g of salt added to it and is stirred until all the salt dissolves	**The mass appears to increase**	**Total mass = mass of salt + mass of water**
A can of lemonade is opened and left for 4 hours	**The mass appears to decrease**	**The lemonade will be flat and some of the gas has escaped**
An ice cube is placed in a beaker and left to melt	**The mass stays the same**	**Nothing has been added or reacted with the ice cube**

